The Pythagorean theorem and its converse

ANSWER KEY

Core Concepts: The Pythagorean theorem and its converse

1 In $\triangle ABC$ studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 88° ✓

b. 98°

c. 80°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 45^\circ) = 88^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

3 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 9

c. 7 ✓

d. 11

Substituting $x = 1$ into $y = 4x + 3$ gives $y = 4(1) + 3 = 7$.

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Procedural Fluency & Algebraic Manipulation

4 In $\triangle ABC$ studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 81° ✓

b. 91°

c. 73°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 52^\circ) = 81^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

6 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 7

b. 13

c. 11 ✓

d. 15

Substituting $x = 2$ into $y = 4x + 3$ gives $y = 4(2) + 3 = 11$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 74° ✓

b. 84°

c. 66°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 59^\circ) = 74^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm ✓

c. 30 cm

d. 96 cm

Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

9 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 11

b. 17

c. 15 ✓

d. 19

Substituting $x = 3$ into $y = 4x + 3$ gives $y = 4(3) + 3 = 15$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 67° ✓

b. 77°

c. 59°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 66^\circ) = 67^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm ✓

c. 36 cm

d. 150 cm

Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

12 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 15

b. 21

c. 19 ✓

d. 23

Substituting $x = 4$ into $y = 4x + 3$ gives $y = 4(4) + 3 = 19$.

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